

LaTeX for Your First Paper

Best Practices and Essential Tools for Mathematical Writing

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Why Choose LaTeX?

- **Professional typesetting:** Beautiful mathematical formulas
- **Consistent formatting:** Focus on content, not appearance
- **Reference management:** Automatic numbering and cross-referencing
- **Publisher ready:** Most journals accept LaTeX submissions
- **Version control friendly:** Plain text format works well with Git

The Goal

Transform your mathematical ideas into publication-ready papers with clarity and elegance.

Basic Document Structure

Every LaTeX document starts with a clear structure:

```
\documentclass{article} % or amsart for math papers

% Essential packages
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage[english]{babel}
\usepackage{mathtools}
\usepackage{amsthm}
\usepackage{amssymb}

\title{Your Paper Title}
\author{Your Name}

\begin{document}
\maketitle
% Your content here
\end{document}
```

Essential Packages for Algebra

`mathtools` Enhanced version of `amsmath`

`amsthm` Theorem environments and proof structure

`amssymb` Additional mathematical symbols

`amsfonts` Mathematical fonts like $\mathbb{R}$, $\mathcal{F}$

Pro Tip

Load `mathtools` instead of `amsmath` - it includes everything from `amsmath` plus useful enhancements.

Math Modes

LaTeX has two main math modes:

- **Inline math:** Use $\$...\$$ or $\backslash(...\backslash)$
- **Display math:** Use $\backslash[...\backslash]$ or `equation` environment

Example

Inline: The order of group G is $|G| = p^n$.

Display:

$$|G| = \sum_{g \in G} 1 = p^n$$

Common Mathematical Notation

Sets and Numbers:

`$\mathbb{N}$`

`$\mathbb{Z}$`

`$\mathbb{Q}$`

`$\mathbb{R}$`

`$\mathbb{C}$`

Operations:

`$\cdot$`

`$\circ$`

`$\times$`

`$\oplus$`

`$\otimes$`

Best Practice

Define custom commands for frequently used symbols:

```
\newcommand{\Z}{\mathbb{Z}}
```

Our Running Example

Throughout this presentation, we'll build a proof of Lagrange's Theorem:

Theorem (Lagrange's Theorem)

Let G be a finite group and H a subgroup of G . Then the order of H divides the order of G .

We'll see how to:

- Structure the theorem statement
- Organize the proof
- Use proper mathematical notation
- Reference definitions and lemmas

Setting Up Theorem Environments

```
\theoremstyle{plain}
\newtheorem{theorem}{Theorem}[section]
\newtheorem{proposition}[theorem]{Proposition}
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}

\theoremstyle{definition}
\newtheorem{definition}[theorem]{Definition}
\newtheorem{example}[theorem]{Example}

\theoremstyle{remark}
\newtheorem{remark}[theorem]{Remark}
```

The `[theorem]` makes all environments share the same counter.

Using Theorem Environments

```
\begin{definition}
```

A $\emph{subgroup}$ H of a group G is a subset $H \subseteq G$ that is closed under the group operation and contains the identity element.

```
\end{definition}
```

```
\begin{lemma}\label{lem:cosets}
```

The left cosets of H in G partition G .

```
\end{lemma}
```

```
\begin{proof}
```

We need to show that every element of G belongs to exactly one left coset...

```
\end{proof}
```

Theorem Environment Styles

Definition (Subgroup)

A *subgroup* H of a group G is a subset $H \subseteq G$ that is closed under the group operation.

Lemma (Coset Partition)

The left cosets of H in G partition G .

Remark

This lemma is crucial for proving Lagrange's theorem.

Note the different formatting: theorems in italic, definitions upright, remarks in smaller font.

Spacing and Punctuation

Good practices:

- Use `\text{...}` for words inside math: x is prime
- Proper spacing with `\,` and `\quad`
- End display equations with punctuation

Example

$$|G| = |H| \cdot [G : H], \quad (1)$$

$$\text{where } [G : H] = \text{number of cosets.} \quad (2)$$

Common Mistake

Don't write: $\$x$ is prime $\$$

Write: “ $\$x \text{ \texttt{is prime}}\$$ ” or “ $\$x\$$ is prime”.

Delimiters and Sizing

Automatic sizing: Use `\left` and `\right`

Bad: $\$(\frac{a}{b})\$$

Good: $\$\left(\frac{a}{b}\right)\$$

Results:

- Bad: $(\frac{a}{b})$
- Good: $\left(\frac{a}{b}\right)$

Manual sizing: `\big`, `\Big`, `\bigg`, `\Bigg`

Create a logical structure:

```
% Preamble: packages and commands

% Document structure:
\section{Introduction}
\section{Preliminary Results}
  \subsection{Group Actions}
  \subsection{Cosets}
\section{Main Results}
  \subsection{Proof of Lagrange's Theorem}
\section{Applications}
\section{Conclusion}
```

Custom Commands and Macros

Define shortcuts for commonly used expressions:

% In preamble:

```
\newcommand{\Z}{\mathbb{Z}}  
\newcommand{\ord}[1]{\lvert#1\rvert} % Order of element  
\newcommand{\normalin}{\triangleleft} % Normal subgroup  
\DeclareMathOperator{\Aut}{Aut} % Automorphism group  
\DeclareMathOperator{\Gal}{Gal} % Galois group
```

Usage: `\ord{G}` produces $|G|$, `\Aut(G)` produces $\text{Aut}(G)$

Why Use Macros?

Consistency, easy global changes, and cleaner source code.

Comments and Documentation

```
% This theorem requires the axiom of choice
\begin{theorem}[Zorn's Lemma]
Every partially ordered set in which every totally
ordered subset has an upper bound contains at least
one maximal element.
\end{theorem}
```

```
% The following proof uses induction on  $|G|$ 
\begin{proof}
We proceed by induction on  $\text{ord}\{G\}$ ...
\end{proof}
```

Comments help you (and collaborators) understand your code later!

Labels and References

```
\begin{theorem}[Lagrange]\label{thm:lagrange}
Let  $G$  be a finite group and  $H$  a subgroup.
Then  $|\mathcal{H}|$  divides  $|\mathcal{G}|$ .
\end{theorem}
```

```
\begin{proof}
By Lemma~\ref{lem:cosets}, the left cosets
partition  $G$ ...
\end{proof}
```

Later in the text: "As we saw in
Theorem~\ref{thm:lagrange}..."

Use descriptive labels: `thm:lagrange`, `lem:cosets`, `eq:main_formula`

Advanced Referencing

```
\begin{equation}\label{eq:order_formula}  
\ord{G} = \ord{H} \cdot [G:H]  
\end{equation}
```

The crucial identity is `equation~\eqref{eq:order_formula}`.

```
\begin{lemma}[Coset Lemma]\label{lem:cosets}  
All left cosets of  $H$  have the same cardinality.  
\end{lemma}
```

By `Lemma~\ref{lem:cosets}`, we conclude that...

Note: `\eqref` automatically adds parentheses around equation numbers.

Step 1: Definitions

```
\begin{definition}[Left Coset]\label{def:coset}  
Let  $H$  be a subgroup of  $G$  and  $g \in G$ .  
The \emph{left coset} of  $H$  containing  $g$  is  
 $gH = \{gh : h \in H\}$ .  
\end{definition}
```

```
\begin{definition}[Index]\label{def:index}  
The \emph{index} of  $H$  in  $G$ , denoted  $[G:H]$ ,  
is the number of distinct left cosets of  $H$  in  $G$ .  
\end{definition}
```

Step 2: Key Lemma

```
\begin{lemma}[Coset Properties]\label{lem:coset_props}
```

Let H be a subgroup of the finite group G . Then:

```
\begin{enumerate}
```

```
\item All left cosets have the same cardinality as  $H$ .
```

```
\item The left cosets of  $H$  partition  $G$ .
```

```
\end{enumerate}
```

```
\end{lemma}
```

```
\begin{proof}
```

```
\emph{Part 1:} The map  $h \mapsto gh$  is a bijection  
from  $H$  to  $gH$ , so  $|gH| = |H|$ .
```

```
\emph{Part 2:} Every  $g \in G$  belongs to the coset  $gH$ .
```

```
Two cosets  $g_1H$  and  $g_2H$  are either identical or disjoint.
```

```
\end{proof}
```

Step 3: Main Theorem

```
\begin{theorem}[Lagrange's Theorem]\label{thm:lagrange}
Let  $G$  be a finite group and  $H$  a subgroup of  $G$ .
Then  $|H|$  divides  $|G|$ .
\end{theorem}
```

```
\begin{proof}
By Lemma~\ref{lem:coset_props}, the left cosets of  $H$ 
partition  $G$ , and each coset has cardinality  $|H|$ .
```

Since there are $[G:H]$ distinct cosets, we have

```
\begin{equation}\label{eq:lagrange}
|G| = |H| \cdot [G:H].
\end{equation}
```

Therefore, $|H|$ divides $|G|$.

```
\end{proof}
```

Introduction to TikZ-CD

For category theory and commutative diagrams:

```
\usepackage{tikz-cd}
```

```
\begin{tikzcd}
```

```
G \arrow[r, "\pi"] \arrow[d, "\phi"]  
  & G/H \arrow[d, "\overline{\phi}"] \\
```

```
K \arrow[r, "\psi"]  
  & K/\ker(\phi)
```

```
\end{tikzcd}
```

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G/H \\ \phi \downarrow & & \downarrow \overline{\phi} \\ K & \xrightarrow{\psi} & K/\ker(\phi) \end{array}$$

Group Homomorphism Diagram

The first isomorphism theorem:

```
\begin{center}
\begin{tikzcd}
G \arrow{rr, "\phi"} \arrow{dr, "\pi"} & & H \\
& G/\ker(\phi) \arrow{ur, "\overline{\phi}"', "\cong"} & 
\end{tikzcd}
\end{center}
```

Results in:

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ & \searrow \pi & \nearrow \overline{\phi} \\ & G/\ker(\phi) & \end{array}$$

Official Documentation:

- `mathtools` package documentation
- `amsthm` user guide
- `tikz-cd` manual

Comprehensive Guides:

- *The Not So Short Introduction to LaTeX* (free PDF)
- *Math into LaTeX* by George Grätzer
- Stack Exchange: tex.stackexchange.com

Online Tools:

- Overleaf (online LaTeX editor)
- Detexify (draw symbols to find LaTeX commands)

Next Steps

Practice with small documents:

- Start with homework solutions
- Type up your notes from algebra courses
- Practice theorem-proof structures

Advanced topics to explore:

- Bibliography management with BibTeX/biblatex
- Custom theorem styles
- Advanced TikZ diagrams
- Beamer presentations (like this one!)

Remember

LaTeX has a learning curve, but the investment pays off with every paper you write.

Questions?

Start your first LaTeX document today!

Happy typesetting!

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